

Labelling Generative AI Assets Using C2PA and JPEG Trust

JPEG Trust Committee

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Abstract

This white paper describes how the C2PA's Technical Specification for Content Credentials and the JPEG Trust specification (ISO/IEC 21617-1) may be used in combination to establish tamper-evident, verifiable, machine-readable provenance labels for assets created or modified by Generative AI systems. The paper examines:

- the complementary roles of both specifications,
- the manifest structures and assertion types relevant to AI-generated content,
- the use of Regions of Interest to locate AI-modified portions of an asset across image, document, and video formats,
- the use of **inputTo** ingredient assertions to record the prompts and reference assets that constitute an AI generation recipe.

Guidance on an end-to-end labelling workflow is provided throughout, with worked examples drawn from multiple asset formats. Trust Profiles and Trust Reports, as defined in JPEG Trust, are addressed in a dedicated annex.

1. Introduction

1.1. Background and Motivation

The rapid proliferation of Generative Artificial Intelligence (Generative AI) tools capable of producing photorealistic images, synthetic audio, and convincing video has created a critical requirement for interoperable, tamper-evident mechanisms to declare and verify the provenance of digital media assets. Without such mechanisms, consumers, platforms, and regulators are unable to reliably distinguish content captured by sensors from content synthesised by AI models — or to identify the extent to which a human-originated asset has subsequently been modified by AI-assisted post-processing.

Two complementary international standards address this challenge:

- **C2PA Technical Specification for Content Credentials** (referenced as *C2PA*), developed by the Coalition for Content Provenance and Authenticity, defines a comprehensive framework for recording the provenance of digital assets through cryptographically signed manifests containing structured assertions.
- **JPEG Trust** (ISO/IEC 21617-1, referenced as *JPEG Trust*), developed by ISO/IEC JTC 1/SC 29/WG 1, extends and profiles the C2PA approach for assets based on ISO/IEC JPEG image standards, introducing the complementary concepts of Trust Indicators, Trust Profiles, and Trust Reports.

Used in combination, these specifications deliver an end-to-end, policy-driven labelling and verification ecosystem that is both human-readable and machine-actionable.

1.2. Scope of This Document

This white paper:

1. Provides an overview of the relevant data structures and assertion types defined in C2PA and JPEG Trust as they pertain to Generative AI content.
2. Describes the normative relationships and points of compatibility between the two specifications.
3. Presents a recommended implementation workflow for labelling assets that are created from scratch by Generative AI, or modified using Generative AI techniques.
4. Describes the use of Regions of Interest to record the spatial, temporal, or textual extents of AI-applied modifications within an asset.
5. Describes the use of **inputTo** ingredient assertions to capture the prompts, reference images, and other inputs that constitute an AI generation recipe.
6. Provides worked examples of manifests and assertions across multiple asset formats.

Trust Profiles and Trust Reports, as defined in JPEG Trust (ISO/IEC 21617-1) and its companion normative annexes, are addressed in [Appendix A, Use of Part 2: Trust Profiles and Reports](#).

This document is informative in nature. In all matters of conformance, the normative requirements of C2PA (specifically §18.2, §18.15, §18.16, §18.28, and §18.29) and JPEG Trust (specifically §4.4, §4.5, §4.6, §4.7, and Annexes A, B, and C) take precedence.

1.3. Normative References

Identifier	Document
Content Credentials	C2PA Technical Specification (v2.4), Coalition for Content Provenance and Authenticity
JPEG Trust	ISO/IEC DIS 21617-1 ed. 2, <i>Information technology — Coding of digital representations of media — Part 1: JPEG Trust</i>
IPTC Digital Source Types	IPTC Digital Source Type NewsCodes
RFC 8949	Bormann, C. and P. Hoffman, "Concise Binary Object Representation (CBOR)", RFC 8949, December 2020
RFC 3161	Adams, C. et al., "Internet X.509 PKI Time-Stamp Protocol (TSP)", RFC 3161, August 2001

2. Conceptual Framework

2.1. The C2PA Provenance Model

The C2PA Technical Specification defines a provenance model centred on three core artifacts [C2PA §2.3]:

Assertion

A data structure representing a statement — either created by the signer or gathered at claim generation time — concerning an asset. Assertions cover areas such as asset creation, edit actions, capture device details, bindings to content, and metadata.

Claim

A digitally signed, tamper-evident data structure that references a set of assertions and the information necessary to represent the content binding [C2PA §2.3.2].

Claim Signature

The digital signature on the claim created using the private key owned by a signer [C2PA §2.3.3].

These artifacts are combined into a **C2PA Manifest** and stored within a **C2PA Manifest Store**, which in non-technical usage is referred to as a **Content Credential** [C2PA §2.3.6].

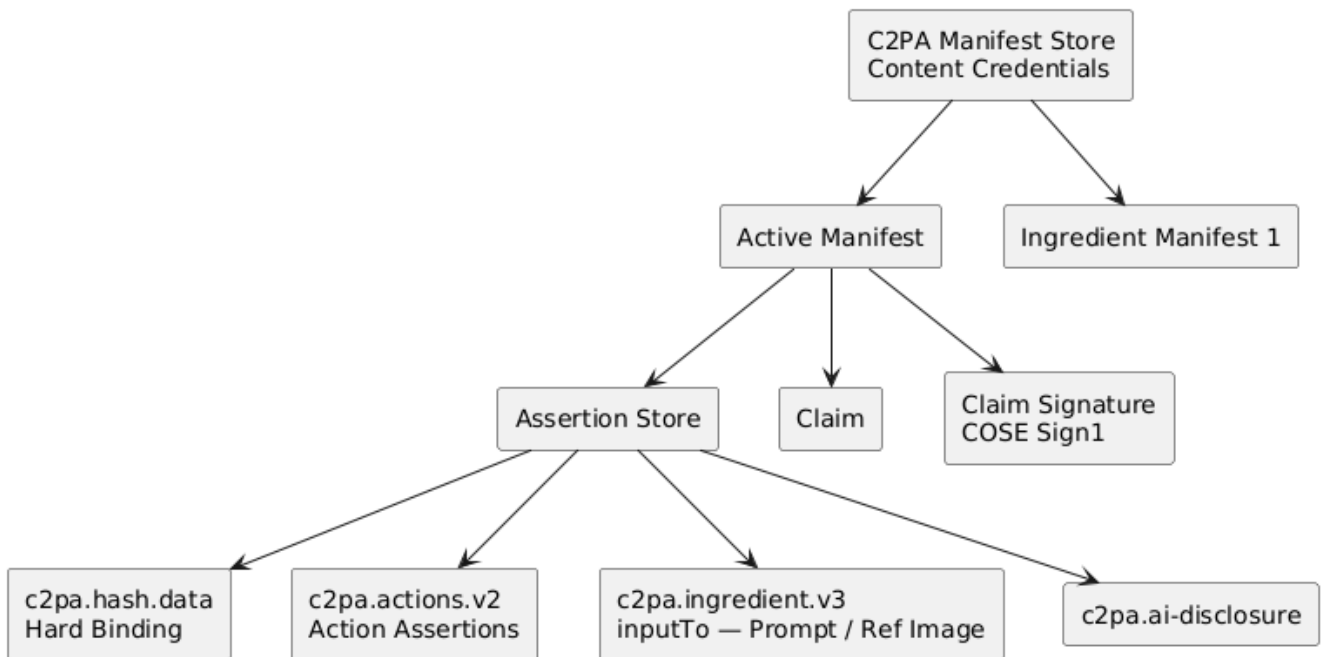


Figure 1. C2PA Manifest Store Structure (derived from C2PA §11.1.4)

The Manifest Store is serialised using the **JPEG Universal Metadata Box Format (JUMBF)** [C2PA §11.1], which provides a hierarchical, typed container format enabling embedding within numerous asset file formats including JPEG, PDF, and video containers.

2.2. The JPEG Trust Framework

JPEG Trust employs the same JUMBF-based building blocks as C2PA, introducing complementary terminology and concepts [JPEG Trust §4.1, Annex E]:

Table 1. Terminology Mapping between JPEG Trust and C2PA (JPEG Trust Annex E, Table E.1)

JPEG Trust Term	C2PA Equivalent
Trust Record	C2PA Manifest Store
Trust Manifest	C2PA Manifest
Trust Declaration	A C2PA Manifest (JUMBF type c2md) with only mandatory assertions; always first in the Trust Record
Media Asset	Asset
Media Asset Provenance	Provenance

The JPEG Trust framework introduces three additional constructs not present in C2PA:

Trust Indicators

Parameters derived from the media asset content, Trust Record, and metadata that may be used to assess the trustworthiness of a media asset in a given context [JPEG Trust §4.5.1].

Trust Profile

A set of expressions evaluated against a Trust Indicator Set to produce a contextualised assessment of trustworthiness [JPEG Trust §4.6].

Trust Report

The result of evaluating a Trust Indicator Set against a Trust Profile [JPEG Trust §4.7.1].

Working examples of these constructs as applied to Generative AI content are provided in [Appendix A, Use of Part 2: Trust Profiles and Reports](#).

2.3. Points of Compatibility

JPEG Trust data structures are built from the same JUMBF building blocks as C2PA [JPEG Trust Annex E §1]. Unless otherwise specified, all C2PA boxes and superboxes are also valid JPEG Trust boxes and superboxes, and may be used interchangeably. The cryptographic primitives — signature schemes and hash algorithms — are identical in both specifications [JPEG Trust Annex E §2].

The primary extension that JPEG Trust introduces is the **Trust Declaration** (**c2md**), a specialised type of manifest that must be the first manifest in any Trust Record and may contain only a specific set of mandatory assertions [JPEG Trust §4.4.4]. While it can be used in a valid Content Credential, it is recommend to not use them as C2PA 2.4 has the same requirements for all manifests.

3. Generative AI Labelling in C2PA

3.1. Standard Manifests and the Inception Action

The primary unit of provenance in C2PA is the **Standard Manifest** (JUMBF type **c2ma**) [C2PA §11.2.2]. Every standard manifest shall contain exactly one hard binding to content assertion and

shall include either a `c2pa.created` or `c2pa.opened` action in its actions assertion [C2PA §15.10.1.2]. For a newly generated Generative AI asset, the appropriate inception action is `c2pa.created`.

NOTE

Within a JPEG Trust workflow, the **Trust Declaration** (`c2md`) fulfils the same role as an initial standard manifest but is additionally constrained to contain only mandatory assertions [JPEG Trust §4.4.4]. The guidance in this section applies equally to both forms. The Trust Declaration specifies that the `c2pa.created` action's `digitalSourceType` field shall be set to `trainedAlgorithmicMedia` for purely AI-generated assets, or `compositeWithTrainedAlgorithmicMedia` where existing media was incorporated as primary input [JPEG Trust §4.4.4 NOTE].

A conformant standard manifest for a Generative AI asset shall contain, at minimum:

1. **Exactly one hard binding assertion** — `c2pa.hash.data`, `c2pa.hash.bboxes`, or one of the other supported options - selected according to asset type [C2PA §11.2.2].
2. **At least one actions assertion** (`c2pa.actions.v2`) containing a `c2pa.created` action specifying:
 - a. The creation timestamp in the `when` field [C2PA §18.15.4.3].
 - b. The `softwareAgent` identifying the Generative AI tool [C2PA §18.15.4.4].
 - c. One of the `digitalSourceType` values from the IPTC vocabulary [C2PA §18.15.4.5], listed in [Section 3.2, “The digitalSourceType Field”](#).

3.2. The `digitalSourceType` Field

3.2.1. General

The `digitalSourceType` field on an action is the primary machine-readable signal for classifying AI-generated content [C2PA §18.15.4.5]. For Generative AI content, the following values are normatively applicable:

Table 2. IPTC `digitalSourceType` Values for Generative AI Content (C2PA §18.15.4.5)

URI	Interpretation
<code>http://cv.iptc.org/newscodes/digitalsourcetype/trainedAlgorithmicMedia</code>	The asset was created entirely by a trained AI/ML model without the direct use of pre-existing media assets as primary inputs. This is the canonical value for pure Generative AI output.
<code>http://cv.iptc.org/newscodes/digitalsourcetype/compositeWithTrainedAlgorithmicMedia</code>	The asset combines AI-generated content with other elements (e.g., a human-originated photograph composited with AI-generated imagery).

3.2.2. `digitalSourceType` on Ingredient Assertions Without a Manifest

A significant use case arises when an ingredient asset — such as a stock photograph used as a reference or composition element — does not carry its own C2PA Manifest. In this situation the claim generator is unable to include an `activeManifest` reference in the ingredient assertion. The `c2pa.ingredient.v3` schema accommodates this through an optional `digitalSourceType` field on the ingredient assertion itself [C2PA §18.16.6, v3 CDDL schema]:

Example 1. Ingredient Assertion with **digitalSourceType** for an asset w/o provenance (C2PA §18.16.6)

```
{
  "dc:title": "Stock Background Photo",
  "dc:format": "image/jpeg",
  "relationship": "componentOf",
  "digitalSourceType":
    "https://cv.iptc.org/newscodes/digitalsourcetype/digitalCapture",
  "description":
    "Stock photo, without a Content Credential, used as background layer; source type
    declared
      by the claim generator as digital capture."
}
```

NOTE The **digitalSourceType** field on an ingredient assertion cannot be combined with an **activeManifest** field — the presence of both constitutes a validation failure [C2PA §15.11.3.2]. Its purpose is specifically to characterise ingredients that lack their own provenance record, preserving the descriptive integrity of the overall provenance chain. This mechanism is particularly relevant in Generative AI workflows where a model may incorporate style-reference assets or training exemplars that predate this ecosystem.

NOTE An additional use case is when an editing tool receives an asset that it knows to have been created by Generative AI, but which does not include the appropriate Content Credential/Trust Record.

3.3. The **c2pa.actions.v2** Assertion for Generative AI

The following example illustrates an actions assertion for a purely AI-generated image (adapted from C2PA §18.15.4.5, Example 8), serialised in CBOR diagnostic notation per RFC 8949 §8:

Example 2. **c2pa.actions.v2** Assertion for a Purely AI-Generated Asset (C2PA §18.15.4.5, Example 8)

```
{
  "actions": [
    {
      "action": "c2pa.created",
      "when": "2025-03-15T14:30:00Z",
      "softwareAgent": {
        "name": "ExampleGenAI Studio",
        "version": "3.1.0",
        "schema.org.SoftwareApplication.operatingSystem": "Linux"
      },
      "digitalSourceType":
        "http://cv.iptc.org/newscodes/digitalsourcetype/trainedAlgorithmicMedia",
      "parameters": {
        "ingredients": [
          {
```

```

        "url": "self#jumbf=c2pa.assertions/c2pa.ingredient.v3",
        "alg": "sha256",
        "hash": "<base64-encoded-hash>"
      }
    ]
  }
}
]
}

```

For an asset originally human-created and subsequently modified using Generative AI (e.g., AI-assisted inpainting), the manifest shall begin with a `c2pa.opened` action referencing the parent ingredient (`parentOf`), followed by one or more additional actions employing the `compositeWithTrainedAlgorithmicMedia` source type [C2PA §18.15.4.2, §18.15.4.5].

3.4. The `c2pa.ai-disclosure` Assertion

C2PA §18.28 introduces a dedicated **AI Disclosure assertion** (`c2pa.ai-disclosure`) to provide machine-readable transparency information beyond that carried by `digitalSourceType`. This assertion provides three critical provenance primitives not previously captured in the ecosystem [C2PA §18.29]:

- **Model provenance** — full disclosure of the AI model(s) used.
- **Scientific routing metadata** — content domain characterisation using the [arXiv](#) taxonomy.
- **Human oversight level** — a structured signal of the degree of human involvement in the generation pipeline.

The `modelType` field is mandatory; all other fields are optional but should be populated where information is available [C2PA §18.28.1.1]:

Example 3. `c2pa.ai-disclosure` Assertion (schema from C2PA §18.29.1)

```

{
  "modelType": "diffusion",
  "modelName": "ExampleDiffusion v4",
  "modelIdentifier": "https://example.org/models/examplediffusion-v4",
  "contentProfile": {
    "humanOversightLevel": "prompt_guided"
  },
  "scientificDomain": ["cs.AI", "cs.CV"]
}

```

3.4.1. Relationship Between `digitalSourceType` and `humanOversightLevel`

C2PA §18.29 explicitly disambiguates the complementary roles of these two fields:

Table 3. Disambiguation of `digitalSourceType` and `humanOversightLevel` (C2PA §18.29)

digitalSourceType	humanOversightLevel	Interpretation
trainedAlgorithmicMedia	fully_autonomous	Pure AI generation; no human review.
trainedAlgorithmicMedia	prompt_guided	AI generation guided by human prompts; moderate human involvement.
compositeWithTrainedAlgorithmicMedia	human_validated	AI edits applied and reviewed or approved by a human editor prior to release.
digitalCreation	not_applicable	No trained model invoked; c2pa.ai-disclosure assertion is not attached.

4. Identifying AI-Modified Regions Using Regions of Interest

4.1. Overview

When Generative AI is applied not to the whole asset but to a specific portion of it — for example, AI inpainting of a face in an image, AI-generated paragraphs inserted into a document, or AI enhancement of a segment of audio — it is both possible and recommended to record the precise extent of that modification. C2PA §18.15.4.6 provides for this through the **changes** field of an action, whose value is an array of **region-map** objects as defined in C2PA §18.2.

The region of interest mechanism supports four orthogonal range types that may be combined within a single region definition [C2PA §18.2.2.1]:

Table 4. Region of Interest Range Types (C2PA §18.2.2)

Range Type	Description
spatial	A physical area of the asset described by a shape (rectangle, circle, or polygon) in pixel or percentage coordinates [C2PA §18.2.2.2].
temporal	A time period described using Normal Play Time (NPT, per RFC 2326) or wall-clock timestamps [C2PA §18.2.2.3].
frame	A range of frames or pages identified by ordinal number, suitable for documents and animations [C2PA §18.2.2.4].
textual	A range of content identified by W3C Web Annotation fragment selectors, supporting structured document formats [C2PA §18.2.2.5].

4.2. Use in Action Assertions

The **changes** field should be included on an action whenever the action was specific to only a portion of the asset [C2PA §18.15.4.6]. For Generative AI modification, this allows a manifest consumer to present a precise spatial, temporal, or textual overlay indicating which regions of an

asset were AI-generated or AI-modified.

The `regionOfInterest` field in assertion metadata [C2PA §18.3.8] additionally enables any assertion — not only an action — to be associated with a region. When an ingredient assertion is relevant only to a specific sub-region of the ingredient asset, the ingredient assertion’s metadata should include a `regionOfInterest` field specifying that sub-region [C2PA §18.16.13].

4.3. Example: JPEG Image — Rectangular Spatial Region (inpainting)

A common Generative AI use case is the inpainting or replacement of a region within an image. The following example records a `c2pa.edited` action with a rectangular spatial region in pixel coordinates describing the area where AI inpainting was applied [C2PA §18.2.4, §18.15.4.6]:

Example 4. Spatial Region of Interest for AI Inpainting in a JPEG Image (C2PA §18.2.4, §18.15.4.6)

```
{
  "actions": [
    {
      "action": "c2pa.opened",
      "when": "2025-04-01T10:00:00Z",
      "softwareAgent": { "name": "ExampleGenAI Studio", "version": "3.1.0" },
      "parameters": {
        "ingredients": [{
          "url": "self#jumbf=c2pa.assertions/c2pa.ingredient.v3",
          "alg": "sha256",
          "hash": "<base64-encoded-hash>"
        }]
      }
    },
    {
      "action": "c2pa.edited",
      "when": "2025-04-01T10:01:30Z",
      "softwareAgent": { "name": "ExampleGenAI Studio", "version": "3.1.0" },
      "digitalSourceType":
        "http://cv.iptc.org/newscodes/digitalsourcetype/compositeWithTrainedAlgorithmicMedia",
      "changes": [
        {
          "region": [
            {
              "type": "spatial",
              "shape": {
                "type": "rectangle",
                "unit": "pixel",
                "origin": { "x": 320.0, "y": 85.0 },
                "width": 415.0,
                "height": 390.0
              }
            }
          ]
        }
      ]
    }
  ]
}
```

```

    ],
    "name": "AI In-painted Region",
    "identifier": "inpaint-background-001",
    "description":
      "Background object replaced using generative AI inpainting"
  }
]
}

```

A validator or display application can use the spatial region data to overlay a visible indicator on the image at coordinates (320, 85) with dimensions 415 x 390 pixels, clearly signalling to users that this area was AI-generated.

4.4. Example: JPEG Image — Polygon Region (AI Object Placement)

Where the AI-generated region is not rectangular, a polygon may be more appropriate. The following example records an AI-generated object placed within an irregularly shaped region of a JPEG image, using percentage coordinates to ensure resolution-independence [C2PA §18.2.2.2]:

Example 5. Polygon Spatial Region for AI-Generated Object Placement in a JPEG (C2PA §18.2.2.2, §18.2.4)

```

{
  "region": [
    {
      "type": "spatial",
      "shape": {
        "type": "polygon",
        "unit": "percent",
        "origin": { "x": 42.0, "y": 18.0 },
        "vertices": [
          { "x": 42.0, "y": 18.0 },
          { "x": 67.0, "y": 15.0 },
          { "x": 78.0, "y": 44.0 },
          { "x": 63.0, "y": 61.0 },
          { "x": 38.0, "y": 55.0 }
        ]
      }
    ]
  },
  "name": "AI-Generated Vehicle",
  "description":
    "Vehicle in foreground generated by AI and composited into scene"
}

```

NOTE

Using "unit": "percent" makes the region definition resolution-independent, which

is preferable where the asset may be re-encoded or scaled downstream.

4.5. Example: PDF Document — Tagged PDF Textual Region

In a PDF document, Generative AI may be used to generate or revise specific paragraphs or sections. For **tagged PDFs**, where the document structure tree is present, textual regions are identified using W3C fragment selector paths into the tag tree [C2PA §18.2.2.5, §18.2.4]:

Example 6. Textual Region of Interest for an AI-Written Section in a Tagged PDF (C2PA §18.2.4, §18.2.2.5)

```
{
  "actions": [
    {
      "action": "c2pa.edited",
      "when": "2025-04-02T09:15:00Z",
      "softwareAgent": {
        "name": "ExampleAI Writer",
        "version": "1.5.0"
      },
      "digitalSourceType":
"http://cv.ipc.org/newscodes/digitalsourcetype/compositeWithTrainedAlgorithmicMedia",
      "changes": [
        {
          "region": [
            {
              "type": "textual",
              "text": {
                "selectors": [
                  [
                    {
                      "selector": {
                        "fragment": "path=/Document/Sect[1]/P[3]"
                      }
                    },
                    {
                      "fragment": "path=/Document/Sect[1]/P[5]"
                    }
                  ]
                ]
              }
            }
          ],
          "name": "AI-Generated Executive Summary",
          "description":
            "Paragraphs 3 to 5 of Section 1 were generated by an AI writing assistant"
        }
      ]
    }
  ]
}
```

```

    }
  ]
}

```

The selector pair `path=/Document/Sect[1]/P[3]` (start) to `path=/Document/Sect[1]/P[5]` (end) addresses paragraphs 3 through 5 of the first section as represented in the PDF tag tree.

4.6. Example: PDF Document — Non-Tagged PDF (Page and Rectangle)

For **non-tagged PDFs**, where no structure tree is available, the textual region may be expressed using a page number and rectangle specification [C2PA §18.2.4]:

Example 7. Textual Region in a Non-Tagged PDF Using Page and Rectangle (C2PA §18.2.4)

```

{
  "region": [
    {
      "type": "textual",
      "text": {
        "selectors": [
          [
            {
              "selector": {
                "fragment": "page=3,rect=72,600,540,750"
              }
            ]
          ]
        ]
      }
    },
    "description": "AI-generated content on page 3, upper body text area"
  ]
}

```

The `page=3,rect=72,600,540,750` fragment identifies page 3 with a bounding rectangle expressed as `left,bottom,right,top` in PDF user space units (points, with origin at the lower-left corner of the page).

4.7. Example: Video — Combined Temporal and Spatial Region

For video assets, AI modification may occur over a defined time window within a defined spatial area — for example, an AI-generated face replacement applied to a specific speaker during a specific segment. Combined temporal and spatial ranges may be used [C2PA §18.2.4]:

```
{
  "region": [
    {
      "type": "temporal",
      "time": {
        "type": "npt",
        "start": "32.5",
        "end": "47.0"
      }
    },
    {
      "type": "spatial",
      "shape": {
        "type": "rectangle",
        "unit": "percent",
        "origin": { "x": 15.0, "y": 5.0 },
        "width": 30.0,
        "height": 45.0
      }
    }
  ],
  "name": "AI Face Replacement – Speaker A",
  "description":
    "Face of Speaker A replaced by AI during seconds 32.5 to 47.0 of the recording"
}
```

5. Capturing the AI Generation Recipe Using **inputTo** Ingredients

5.1. Overview

A key requirement for AI provenance is the ability to record not just *what* was generated, but *how* it was instructed. The generation recipe — comprising the text prompts, negative prompts, seed values, reference images, style exemplars, and any other inputs provided to the AI system — constitutes essential provenance information for understanding the origin of an AI-generated asset.

C2PA provides a direct mechanism for this through the **inputTo** relationship value of the `c2pa.ingredient.v3` assertion [C2PA §18.16.3, Table 10]:

inputTo: This ingredient was used as input to a computational process, such as an AI/ML model, that led to the creation or modification of this asset.

The **inputTo** relationship is explicitly designed to represent inputs to AI/ML inference processes,

including both training dataset references and inference-time inputs [C2PA §18.16.1]. Unlike `parentOf` (which requires `c2pa.opened`) or `componentOf` (which requires `c2pa.placed`), the `inputTo` relationship carries no mandatory associated action in the `c2pa.actions.v2` assertion [C2PA §18.16.3]. The AI inference process itself produces the output; the inputs are contextual provenance rather than compositional derivation.

NOTE

During ingredient validation, a validator encountering an `inputTo` ingredient that lacks an `activeManifest` field will record `ingredient.unknownProvenance` rather than treating the absence as a validation failure [C2PA §15.11.3.3, validation flowchart]. This reflects the reality that many AI inputs — raw text prompts, hyperparameter sets — do not have associated C2PA manifests.

5.2. Representing Text Prompts as Ingredients

A text prompt used to instruct a Generative AI model may be embedded directly within the manifest using an **Embedded Data assertion** referenced from the ingredient's `data` field [C2PA §18.12, §18.16.6]:

Example 9. `inputTo` Ingredient Assertion for a Text Prompt (C2PA §18.16.3, §18.16.6)

```
{
  "dc:title": "Generation Prompt",
  "dc:format": "text/plain",
  "relationship": "inputTo",
  "description":
    "Text prompt provided to the diffusion model for this generation",
  "data": {
    "url": "self#jumbf=c2pa.assertions/c2pa.embedded-data",
    "alg": "sha256",
    "hash": "<base64-encoded-hash-of-embedded-data>"
  },
  "dataTypes": [
    {
      "type": "c2pa.types.generator.prompt"
    }
  ]
}
```

The `dataTypes` field provides typed characterisation of the embedded data, enabling manifest consumers to identify prompt data without parsing its content [C2PA §18.16.5].

5.3. Representing Reference Images as Ingredients

Where a reference image (e.g., a style exemplar, a seed image, or a control image for a ControlNet-style model) was used as input to the AI inference process, it should be recorded as an `inputTo` ingredient. If the reference image carries its own C2PA Manifest, the `activeManifest` and `claimSignature` fields shall be included to preserve the provenance chain:

```
{
  "dc:title": "Style Reference – Watercolour Exemplar",
  "dc:format": "image/jpeg",
  "relationship": "inputTo",
  "description":
    "Reference image used to guide the diffusion model's visual style",
  "activeManifest": {
    "url":
      "self#jumbf=c2pa/urn:c2pa:A3F7-0012-4E9B-BBCD-001122334455/c2pa.claim.v2",
    "alg": "sha256",
    "hash": "<base64-encoded-hash>"
  },
  "claimSignature": {
    "url":
      "self#jumbf=c2pa/urn:c2pa:A3F7-0012-4E9B-BBCD-001122334455/c2pa.signature",
    "alg": "sha256",
    "hash": "<base64-encoded-hash>"
  }
}
```

If the reference image does not carry a C2PA Manifest, the **digitalSourceType** field shall be used on the ingredient assertion to characterise its source, as described in [Section 3.2](#), “[The digitalSourceType Field](#)”.

5.4. Representing Multiple Inputs — A Complete Recipe

A real Generative AI workflow typically involves multiple distinct inputs: a positive prompt, a negative prompt, seed and hyperparameter values, a style reference image, and possibly an inpainting mask. Each should be recorded as a separate **inputTo** ingredient assertion, labelled according to the standard multiple-instance labelling rules [C2PA §18.16.1 NOTE]:

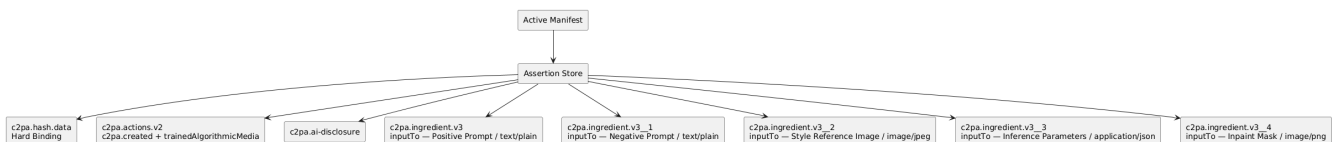


Figure 2. Multiple **inputTo** Ingredient Assertions Representing a Generation Recipe (C2PA §18.16.1)

All ingredient assertions shall be referenced from the **parameters.ingredients** array of the **c2pa.created** action [C2PA §18.15.4.7]:

```
{
  "actions": [
    {
      "action": "c2pa.created",
```



```

"when": "2025-04-05T16:00:00Z",
"softwareAgent": {
  "name": "ExampleGenAI Studio",
  "version": "3.1.0"
},
"digitalSourceType":
  "http://cv.ip tc.org/news codes/digital sourcetype/trainedAlgorithmicMedia",
"parameters": {
  "ingredients": [
    {
      "url": "self#jumbf=c2pa.assertions/c2pa.ingredient.v3",
      "alg": "sha256",
      "hash": "<hash-of-positive-prompt-ingredient>"
    },
    {
      "url": "self#jumbf=c2pa.assertions/c2pa.ingredient.v3__1",
      "alg": "sha256",
      "hash": "<hash-of-negative-prompt-ingredient>"
    },
    {
      "url": "self#jumbf=c2pa.assertions/c2pa.ingredient.v3__2",
      "alg": "sha256",
      "hash": "<hash-of-style-reference-ingredient>"
    },
    {
      "url": "self#jumbf=c2pa.assertions/c2pa.ingredient.v3__3",
      "alg": "sha256",
      "hash": "<hash-of-inference-params-ingredient>"
    },
    {
      "url": "self#jumbf=c2pa.assertions/c2pa.ingredient.v3__4",
      "alg": "sha256",
      "hash": "<hash-of-inpaint-mask-ingredient>"
    }
  ]
}
}
]
}

```

5.5. Representing Non-Media Inputs — Inference Hyperparameters

Scalar values and structured parameters (e.g., random seed, guidance scale, number of inference steps) that do not have a natural file-based representation may be embedded as JSON in an embedded data assertion and referenced as an `inputTo` ingredient:

```
{
  "dc:title": "Inference Parameters",
  "dc:format": "application/json",
  "relationship": "inputTo",
  "description": "Inference hyperparameters used for this generation run",
  "data": {
    "url": "self#jumbf=c2pa.assertions/c2pa.embedded-data__1",
    "alg": "sha256",
    "hash": "<base64-encoded-hash>"
  },
  "dataTypes": [
    {
      "type": "c2pa.types.generator.parameters"
    }
  ]
}
```

The corresponding embedded data assertion would carry a JSON object such as:

```
{
  "seed": 2847391052,
  "guidance_scale": 7.5,
  "num_inference_steps": 50,
  "scheduler": "DPMSolverMultistepScheduler"
}
```

5.6. Training Data References

In addition to inference-time inputs, the **inputTo** relationship may be used to reference the dataset used to train the AI model. C2PA §9.2.5 describes the **c2pa.hash.collection.data** assertion specifically for use with collections such as AI/ML training datasets, noting that this assertion can describe each folder of a training dataset for an AI/ML model [C2PA §9.2.5 NOTE]. This enables verifiable, hash-bound association between the generated asset and the training corpus.

6. End-to-End Implementation Workflow

6.1. Overview

The following sequence describes a complete workflow for labelling a Generative AI asset, integrating the manifest structure, regions of interest, and **inputTo** ingredients described in the preceding sections.

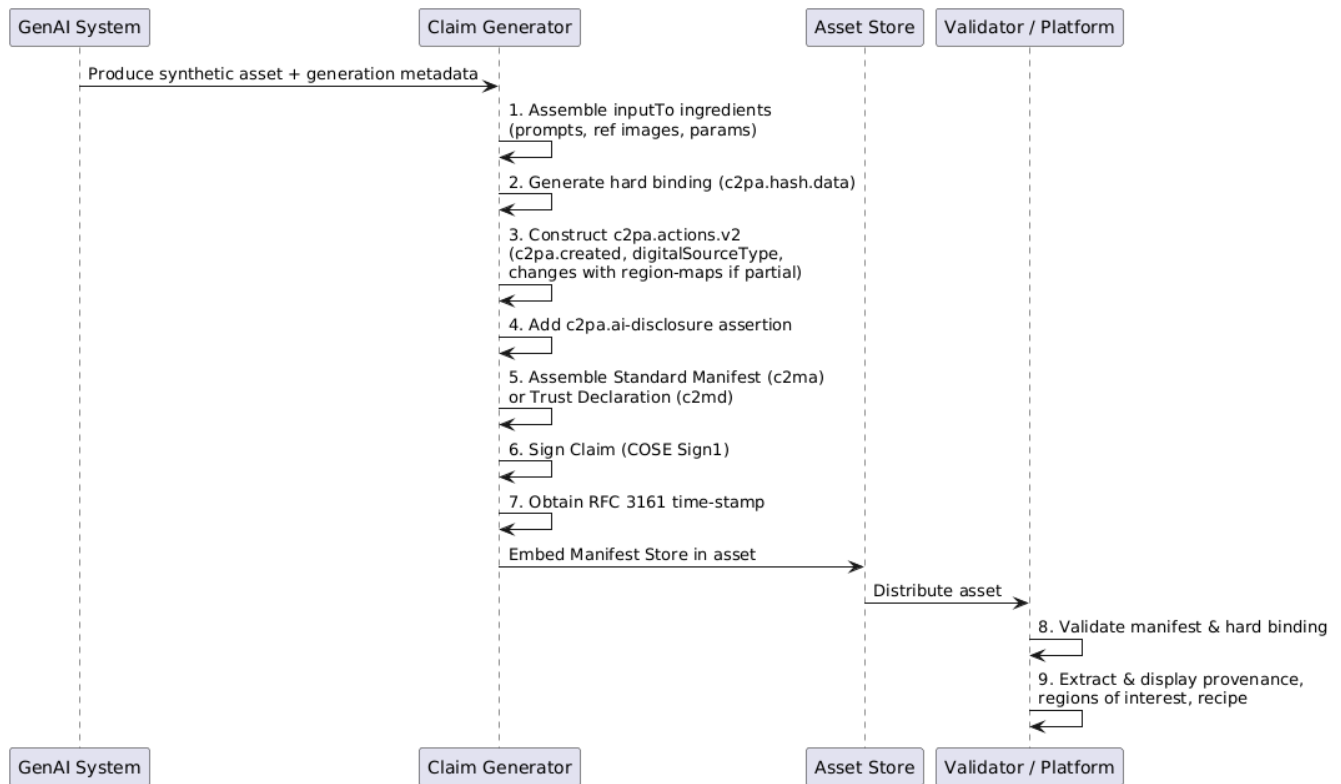


Figure 3. End-to-End Generative AI Labelling Workflow

6.2. Phase 1 — Asset Creation

6.2.1. Step 1: Assemble **inputTo** Ingredients

Before computing the hard binding, the claim generator constructs ingredient assertions for each input to the AI generation process: prompts, reference images, and parameter sets. Each ingredient assertion uses the **inputTo** relationship. Embedded data assertions are created to hold text-format inputs (prompts, JSON parameters).

6.2.2. Step 2: Generate the Hard Binding

The claim generator computes a content hash using the appropriate binding assertion type for the asset format [C2PA §9.2, §11.2.2]. For JPEG assets, **c2pa.hash.data** with exclusion ranges for the manifest placeholder are used [C2PA §10.4].

6.2.3. Step 3: Construct the Actions Assertion

The **c2pa.actions.v2** assertion shall contain a **c2pa.created** action with:

- The **when** timestamp [C2PA §18.15.4.3].
- The **softwareAgent** identifying the AI tool [C2PA §18.15.4.4].
- The appropriate **digitalSourceType** value [C2PA §18.15.4.5].
- A **changes** field containing **region-map** objects for any partial AI modification [C2PA §18.15.4.6].
- A **parameters.ingredients** array referencing any **inputTo** ingredient assertions [C2PA §18.15.4.7].

6.2.4. Step 4: Add the AI Disclosure Assertion

A `c2pa.ai-disclosure` assertion should be added, with at minimum the `modelType` field populated. The `humanOversightLevel` in `contentProfile` should accurately reflect the generation pipeline [C2PA §18.28, §18.29].

6.2.5. Step 5: Assemble and Sign the Manifest

The manifest shall be assembled and signed using the signer's X.509 credential. A trusted RFC 3161 time-stamp should be obtained and incorporated [C2PA §10.3.2.4, §10.3.2.5].

6.2.6. Step 6: Embed the Trust Record

The Trust Record shall be embedded into the asset using the appropriate method for the format [JPEG Trust §6.2, C2PA Annex A]:

- **JPEG 1 / JPEG XT**: APP11 marker segment.
- **JPEG XL**: `jumb` box within the JXL container.
- **JPEG 2000**: UUID box within the JP2/JPX container.
- **Video (BMFF-based)**: BMFF-compatible embedding per C2PA Annex A.
- **other formats***: As per C2PA embedding guidance.

6.3. Phase 2 — Subsequent Modification

If a Generative AI asset undergoes further modification, the modifier shall create a new standard manifest that:

1. References the prior active manifest as a `parentOf` ingredient with a corresponding `c2pa.opened` action.
2. Records modification actions in a new `c2pa.actions.v2` assertion, with `changes` region-maps where applicable.
3. Computes a new hard binding over the modified asset.

The new manifest is appended to the existing Manifest Store, preserving the complete provenance chain [C2PA §10.3.2.2].

7. Conclusion

The C2PA Technical Specification and JPEG Trust (ISO/IEC 21617-1) provide complementary and mutually reinforcing mechanisms for the verifiable labelling of Generative AI assets. C2PA provides the foundational data structures — manifests, assertions, claims, and cryptographic bindings — while JPEG Trust builds upon these to define a policy-driven evaluation layer through Trust Indicators, Trust Profiles, and Trust Reports.

The key integration points described in this paper are:

1. **Standard Manifests** which mandate the `c2pa.created` inception action with `digitalSourceType` as the primary machine-readable AI creation classification signal along with `c2pa.opened` for subsequent modifications. It also accommodates `digitalSourceType` on ingredient assertions for un-manifested source assets.
2. The **`c2pa.ai-disclosure` assertion**, which provides granular model provenance, scientific domain classification, and human oversight level signals complementing the `digitalSourceType` classification.
3. **Regions of Interest** in the `changes` field of action assertions, enabling precise spatial, temporal, and textual localisation of AI-modified portions of an asset across image, document, and video formats.
4. **inputTo ingredient assertions** (`c2pa.ingredient.v3`), which provide a normative, verifiable mechanism for capturing the complete AI generation recipe — prompts, reference assets, seed values, and model parameters — as cryptographically bound provenance data.
5. The JPEG Trust **Trust Profile and Trust Report** mechanism, which enables platform operators and regulators to define and apply context-specific AI content policies in an interoperable, machine-actionable manner (see [Appendix A, Use of Part 2: Trust Profiles and Reports](#)).

Together, these specifications provide the technical foundation for a trustworthy, scalable, and interoperable ecosystem for Generative AI provenance disclosure.

Appendix A: Use of Part 2: Trust Profiles and Reports

This annex addresses the Trust Indicator Set, Trust Profile, and Trust Report constructs introduced by JPEG Trust (ISO/IEC 21617-1) and its normative Annexes A, B, and C. These constructs provide the policy evaluation layer that operates over the provenance data recorded in JPEG Trust Records (Content Credentials).

A.1. Trust Indicator Sets for AI-Generated Content

A.1.1. Overview

A **Trust Indicator Set (TIS)** is a JSON-LD document aggregating Trust Indicators extracted from the Trust Record, asset metadata, and asset content [JPEG Trust §4.5.2, Annex A]. For a purely AI-generated asset, the TIS's `declaration` block reflects the assertions from the first (Trust Declaration or equivalent) manifest.

A.1.2. Trust Report Generation Procedure

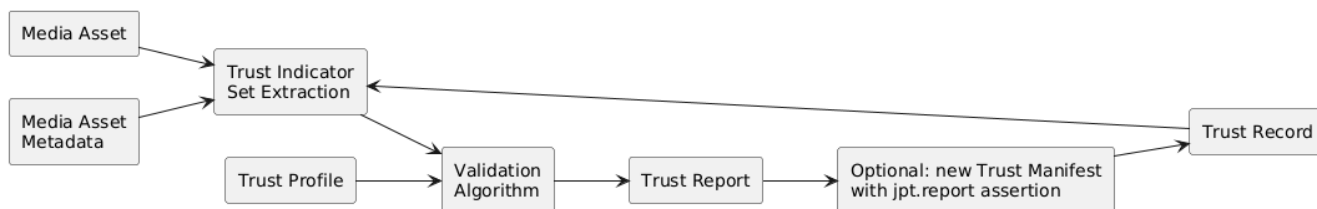


Figure 4. Trust Report Generation Procedure (derived from JPEG Trust §4.7.2)

A.1.3. Example Trust Indicator Set

The following example (adapted from JPEG Trust Annex A, Figure A.13) illustrates a TIS for a purely AI-generated PNG image:

Figure A-2. Trust Indicator Set for a Generative AI Asset (adapted from JPEG Trust Annex A, Figure A.13)

```

{
  "@context": ["https://jpeg.org/jpegtrust/"],
  "declaration": {
    "claim.v2": {
      "alg": "sha256",
      "claim_generator": "ExampleGenAI Studio/3.1.0 (Linux)",
      "claim_generator_info": [
        {
          "name": "ExampleGenAI Studio",
          "version": "3.1.0",
          "schema.org.SoftwareApplication.operatingSystem": "Linux"
        }
      ],
      "signature": {
        "self#jumbf=c2pa/urn:c2pa:5560f310-addc-4823-bce7-619e070fafa9/c2pa.signature",
        "dc:format": "image/png",
        "signature_status": "claimSignature.validated",
        "assertion_status": {
          "c2pa.hash.data": "assertion.hashedURI.match",
          "c2pa.actions.v2": "assertion.hashedURI.match"
        },
        "content_status": "assertion.dataHash.match"
      },
      "assertions": {
        "c2pa.hash.data": {
          "alg": "sha256",
          "hash": "<base64-encoded-hash>",
          "exclusions": [{ "start": 9960, "length": 4213 }]
        },
        "c2pa.actions.v2": {
          "actions": [
            {
              "action": "c2pa.created",
              "when": "2025-03-15T14:30:00Z",
              "softwareAgent": { "name": "ExampleGenAI Studio" },

```

```

      "digitalSourceType":
        "https://cv.iptc.org/newscodes/digitalsourcetype/trainedAlgorithmicMedia"
      }
    ]
  }
}
}
}

```

A.2. Trust Profiles for Generative AI Classification

A.2.1. Design Principles

A **Trust Profile** defines the expressions to be evaluated against a Trust Indicator Set [JPEG Trust §4.6, Annex B]. Trust Profile expressions use the **json-formula** language. The output of each expression statement is both recorded in the Trust Report and written back into the TIS under the **profile** namespace, enabling subsequent statements to reference earlier results [JPEG Trust Annex B].

For AI content labelling, a Trust Profile shall at minimum: verify the content binding integrity; determine whether the asset was produced by Generative AI via **digitalSourceType**; assess whether the asset was modified after initial creation; and evaluate overall compliance.

A.2.2. Example Trust Profile

The following Trust Profile (adapted from JPEG Trust Annex B, Figure B.8) classifies and reports on Generative AI assets:

Figure A-3. Trust Profile for Generative AI Asset Classification (adapted from JPEG Trust Annex B, Figure B.8)

```

---
metadata:
  name: Generative AI Content Profile
  issuer: Example Standards Body
  date: 2025-04-01T00:00:00.000Z
  version: 1.0.0
  language: en

---
# Section 1 – General Integrity

- id: generalInfo
  title: General Information
  report_text: This section assesses the general integrity of the media asset.

- id: content
  description: Verifies the hard binding.

```

```

expression: >
  declaration.'claim.v2'.content_status == "assertion.dataHash.match"
report_text:
  "true":
    en: The content binding is intact; this asset has not been tampered with.
  "false":
    en: The content binding check failed; the asset may have been modified
        without a corresponding provenance record.

```

Section 2 – Generative AI Classification

- id: genAI
 - title: Generative AI Usage
 - report_text: >

This section provides information about whether the media asset was produced by Generative AI.
- id: aigc
 - description: >

Matches both trainedAlgorithmicMedia and compositeWithTrainedAlgorithmicMedia.
 - expression: >


```
contains(
  declaration.assertions.'c2pa.actions.v2'.actions[0].digitalSourceType,
  "trainedAlgorithmicMedia"
)
```
 - report_text:
 - "true":
 - en: This media asset was produced by Generative AI.
 - "false":
 - en: This media asset was not produced by Generative AI.
- id: declaration_only
 - description: Checks whether additional Trust Manifests are present.
 - expression: >


```
contains(keys(@), "declaration") && !contains(keys(@), "manifests")
```
 - report_text:
 - "true":
 - en: No modifications have been recorded after initial creation.
 - "false":
 - en: >

This media asset was modified after initial Generative AI creation; a full provenance chain is present.

Section 3 – Compliance

- id: compliance
 - title: Compliance
 - report_text: >

This section provides the compliance status of the media asset with respect to this profile.

```
- id: jpt:profile_compliance
  expression: >
    @.profile.aigc && (@.profile.declaration_only || @.profile.content)
  report_text:
    "true":
      en: This media asset is compliant with the Generative AI Content Profile.
    "false":
      en: This media asset is not compliant with the Generative AI Content Profile.
```

NOTE

The `aigc` expression uses `contains()` so that it correctly matches both `trainedAlgorithmicMedia` and `compositeWithTrainedAlgorithmicMedia` values, as recommended in JPEG Trust Annex B §B.3.

A.3. Trust Reports for Generative AI Assets

A.3.1. Example Trust Report

The following example (adapted from JPEG Trust Annex C, Figure C.3) illustrates the Trust Report produced by evaluating the TIS from Figure A-2 against the Trust Profile from Figure A-3:

Figure A-4. Trust Report for a Generative AI Asset (adapted from JPEG Trust Annex C, Figure C.3)

```
profile_metadata:
  name: Generative AI Content Profile
  issuer: Example Standards Body
  date: 2025-04-01T00:00:00.000Z
  version: 1.0.0
  language: en

sections:
  - - id: generalInfo
      title: General Information
      report_text: This section assesses the general integrity of the media asset.
    - id: content
      value: true
      report_text: The content binding is intact; this asset has not been tampered
with.

    - - id: genAI
        title: Generative AI Usage
        report_text: >
          This section provides information about whether the media asset was
          produced by Generative AI.
      - id: aigc
        value: true
        report_text: This media asset was produced by Generative AI.
```

- id: declaration_only
value: true
report_text: No modifications have been recorded after initial creation.
- - id: compliance
title: Compliance
report_text: >
This section provides the compliance status of the media asset with respect to this profile.
- id: jpt:profile_compliance
value: true
report_text: >
This media asset is compliant with the Generative AI Content Profile.

A.3.2. Embedding the Trust Report

A Trust Report may be embedded back into the asset as a `jpt.report` assertion within an **Update Manifest** (not a new Trust Declaration, since no content modification is being performed) [JPEG Trust §4.7.3, C2PA §11.2.3]. The Trust Report assertion shall have the label `jpt.report` [JPEG Trust §4.7.3].

Appendix B: Abbreviations

Abbreviation	Expansion
AI	Artificial Intelligence
AIGC	AI-Generated Content
CBOR	Concise Binary Object Representation (RFC 8949)
C2PA	Coalition for Content Provenance and Authenticity
COSE	CBOR Object Signing and Encryption
IPTC	International Press Telecommunications Council
JUMBF	JPEG Universal Metadata Box Format
ML	Machine Learning
NPT	Normal Play Time (RFC 2326)
ROI	Region of Interest
TIS	Trust Indicator Set
TSA	Time Stamping Authority (RFC 3161)
XMP	Extensible Metadata Platform